

A bad "undecidability" proof.

Let $L = \{ \langle D, x \rangle \mid D \text{ is a DFA and } D \text{ accepts } x \}$.

"Theorem" L is undecidable.

Proof: Suppose not. Let O be a decider for L . Let $\langle M, w \rangle$ be an instance of HALT.

error!
takes a DFA

$M' =$ "On input x :

- Run M on w .
- If we reach this point, that means M halted on w . So, we accept x ."

not a DFA!
If it was, it cannot encode $\langle M, w \rangle$ (a DFA always halts for a given string)

We run M' as input to O .
If O accepts, then M halted on w .
Otherwise, M did not halt on w .
We have decided $\langle M, w \rangle$, so we can decide HALT, a contradiction. $\square$

